

Jaundel Simon

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EDUCATION

Toronto Metropolitan University

Toronto, ON

Bachelor of Engineering in Computer Engineering; GPA: 3.38/4.33

Sep. 2023 – May 2028

- **Relevant Coursework:** Computer Architecture, Digital Systems, Data Structures & Algorithms, Microprocessor Systems, Object-Oriented Design

TECHNICAL SKILLS

Languages: Python, Java, C, Assembly (HCS12), VHDL, MATLAB

Developer Tools: Git, Github, VS Code, Intel Quartus Prime, ModelSim, PyCharm, CodeWarrior, NetBeans

PROJECTS

SEVEN (Sustainable Energy via Efficient Neural-routing) | *Python, Lemonade Server*

Nov. 2025

- Founded SEVEN, an AI routing system that cuts inference energy by 38% vs. cloud-only by routing 40% of queries to local SLMs; led 5-person team to win TMU Hack the World (\$1,625) with prototype projecting 483 GWh annual savings (UN SDG 7).
- Built an adaptive routing framework to delegate simple prompts to local SLMs and complex queries to the cloud, achieving 90% accuracy (human-validated) and 94% energy reduction per local query using zero-cost heuristics, prompt engineering, and API augmentation.
- Designed a model-agnostic Lemonade Server adapter with runtime-configurable CPU/GPU/NPU selection that surfaces per-call energy and Wh-saved metrics, using exponential-backoff retries and local/cloud fallback to keep inference resilient under model or network failures.
- Authored runbooks to guide initial setup and managed PRs/code reviews to support front-end TUI development, energy profiling, and SQL-backed conversation/energy logging.

Neural Processing Unit (NPU) Core | *VHDL, Intel Quartus Prime, ModelSim*

Sep 2025 – Oct 2025

- Architected a neuron accelerator computing $y = \text{ReLU}(W_1 \cdot A + W_2 \cdot B + \text{BIAS})$ with 24-cycle deterministic latency and 2M inferences/sec throughput for single-neuron inference workload using 8-bit signed fixed-point arithmetic.
- Implemented modular datapath with an 8-cycle sequential signed multiplier (shift-add), saturating accumulator, and combinational ReLU; closed timing at 182 MHz on Cyclone V.
- Engineered 11-state FSM controller coordinating dual MAC operations and bias accumulation with ready-valid protocol; integrated performance counters for cycle-accurate profiling.
- Verified complete RTL functionality using self-checking ModelSim testbench covering edge cases (overflow, saturation, sign extension).

MNIST Neural Network (NumPy & PyTorch) | *Python, NumPy, PyTorch*

Aug 2025 – Sep 2025

- Developed handwritten digit classifiers achieving 99.05% accuracy (PyTorch CNN) vs 97.59% (NumPy MLP), quantifying 1.46pp accuracy gain against 3.4× inference latency cost (0.33ms→1.10ms) on 10K samples.
- Constructed a fully vectorized NumPy MLP in 60 lines without autograd, implementing manual forward/backpropagation while building a parallel PyTorch CNN with convolutional layers on identical 60K samples.
- Executed systematic 70-experiment grid search across 22 configurations with deterministic seeding, validating robustness with 14 setups exceeding 98% accuracy and capturing negative controls.
- Automated end-to-end benchmarking with ONNX export and CSV metric logging, creating auditable experiment history for deployment validation and regression tracking.

FPGA-Deployed 8-Bit CPU | *VHDL, Intel Quartus Prime, ModelSim, Altera DE2*

Nov 2024 – Dec 2024

- Architected and integrated a complete 8-bit CPU from 19 synthesizable VHDL modules, deploying three distinct system builds to an Altera DE2 board with validated switch and display I/O.
- Achieved timing closure for the board's native 50 MHz clock using TimeQuest static timing analysis, verifying cycle-accurate execution for all single-cycle and multi-cycle operations.
- Devised the core datapath with a multi-function ALU supporting 8 opcode-driven operations and an 8-state FSM controller to enable robust, multi-cycle instruction sequencing.
- Implemented a rigorous verification flow from RTL simulation to on-board validation, ensuring first-pass hardware success and passing all functional test vectors.